

KANCHANA RANASINGHE

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EDUCATION

Stony Brook University, NY, USA

PhD in Computer Science

Thesis: *Connecting Natural Language to Video with Minimal Human Supervision*

Aug 2021 - Feb 2026

University of Moratuwa, Sri Lanka

BSc in Engineering; Awarded Most Outstanding Graduand of the Year

Senior Thesis: *Realtime Multi-Object Tracking and Pixelwise Segmentation*

Dec 2015 - Jan 2020

RESEARCH EXPERIENCE

Salesforce AI Research, NY, USA

Research Scientist

Research Intern

March 2026 - Present

June 2025 - Jan 2026

- Learning Language-Motion Relations from Web-Scale Human Activity Videos
- Unified Generative Models for Language Conditioned Robot Control

Google Research, NYC, USA - Student Researcher

March 2024 - Dec 2024

- Continuous-space CRF formulation for modeling natural images
- Integrating CRF with latent diffusion for efficient image generation

Meta AI Research, NYC, USA - Research Scientist Intern

May 2023 - Dec 2023

- Spatial reasoning in multi-modal large language models
- Localization awareness in video-language models

Apple MLR, Cupertino, USA - Machine Learning Research Intern

May 2022 - Sep 2022

- Multi-modal self-supervised representation learning
- Interpretability and robustness of vision language models

MBZUAI, Abu Dhabi, UAE - Research Assistant

Nov 2020 - Aug 2021

- Representation learning: contrastive losses, self-supervised objectives
- Interpretability, robustness, and adversarial attacks for vision transformers

Wenn ASA, Stavanger, Norway - Machine Learning Engineer (remote)

Jan 2019 - Oct 2020

- Leading team of three associate data scientists for vehicle damage detection project
- Focus on efficient mobile implementations, instance segmentation models, and active learning

FiveAI, Cambridge, UK - Research Intern

June 2018 - Dec 2018

- Perception team of self-driving startup
- 3D orientation estimation: improve occluded object handling in videos with synthetic data

SELECTED AWARDS

Outstanding Demo Award at CVPR 2023

Burgert, Ranasinghe, Li, Ryoo, “*Diffusion Illusions: Hiding Images in Plain Sight*”

Most Outstanding Graduand of the Year

Class of 2020 at University of Moratuwa, Sri Lanka

- Future Optical Flow Prediction Improves Robot Control & Video Generation** CVPR Findings, 2026
K Ranasinghe, H Zhou, Y Fang, L Yang, L Xue, R Xu, C Xiong, S Savarese, MS Ryoo, JC Niebles
- Pixel Motion Diffusion is What We Need for Robot Control** CVPR, 2026
ER Nguyen, Y Zhang, K Ranasinghe, X Li, MS Ryoo
- IVRA: Improving Visual-Token Relations for Robot Action Policy with Training-Free Hint-Based Guidance** ICRA, 2026
J Park, K Ranasinghe, J Jang, C Mata, YS Jang, MS Ryoo
- Too Many Frames, not all Useful: Efficient Strategies for Long-Form Video QA** EACL, 2026
J Park, K Ranasinghe, K Kahatapitiya, W Ryoo, D Kim, M S Ryoo
- Understanding Long Videos with Multimodal Language Models** ICLR, 2025
K Ranasinghe, X Li, K Kahatapitiya, M S Ryoo
- LLaRA: Supercharging Robot Learning Data for Vision-Language Policy** ICLR, 2025
X Li, C Mata, J Park, K Kahatapitiya, Y Jang, J Shang, K Ranasinghe, R Burgert, M Cai, Y Lee, M S Ryoo
- Language Repository for Long Video Understanding** ACL Findings, 2025
K Kahatapitiya, K Ranasinghe, J Park, M S Ryoo
- Unsupervised Domain Adaptive Segmentation using Domain-Agnostic Text** ECCV, 2024
C Mata, K Ranasinghe, M S Ryoo
- Learning to Localize Objects Improves Spatial Reasoning in Visual-LLMs** CVPR, 2024
K Ranasinghe, S N Shukla, O Poursaeed, M S Ryoo, T Y Lin
- Diffusion Illusions: Hiding Images in Plain Sight** SIGGRAPH, 2024
R Burgert, X Li, A Leite, K Ranasinghe, M Ryoo
- Hierarchical Text-to-Vision Self Supervised Alignment for Improved Histopathology Representation Learning** MICCAI 2024
H Watawana, K Ranasinghe, T Mahmood, M Naseer, S Khan, F S Khan
- Language-based Action Concept Spaces Improve Video SSL** NeurIPS, 2023
K Ranasinghe, M S Ryoo
- Perceptual Grouping in Contrastive Vision-Language Models** ICCV, 2023
K Ranasinghe, B McKinzie, S Ravi, Y Yang, A Toshev, J Shlens
- Peekaboo: Text to Image Diffusion Models are Zero-Shot Segmentors** CVPR workshop, 2023
R Burgert, K Ranasinghe, X Li, M S Ryoo
- Self-supervised Video Transformers** CVPR, 2022 (oral)
K Ranasinghe, M Naseer, S Khan, F Khan, M S Ryoo
- On Improving Adversarial Transferability of Vision Transformers** ICLR, 2022 (spotlight)
M Naseer, K Ranasinghe, S Khan, F Khan, F Porikli
- Intriguing Properties of Vision Transformers** NeurIPS, 2021 (spotlight)
M Naseer, K Ranasinghe, S Khan, M Hayat, F Khan, M Yang
(*listed among top 15 most cited papers at NeurIPS 2021* - [link](#))
- Orthogonal Projection Loss** ICCV, 2021
K Ranasinghe, M Naseer, M Hayat, S Khan, F Khan
- Conditional Generative Modeling via Learning the Latent Space** ICLR, 2021
S Ramasinghe, K Ranasinghe, S Khan, N Barnes, S Gould
- Bipartite Conditional Random Fields for Panoptic Segmentation** BMVC, 2020 (oral)
S. Jayasumana, K Ranasinghe, M. Jayawardhana, S. Liyanaarachchi and H. Ranasinghe
- Micro Actions and Deep Static Features for Activity Recognition** DICTA, 2017
S. Ramasinghe, J. Rajasegaran, V. Jayasundara, K. Ranasinghe, R. Rodrigo and A. Pasqual

PUBLICATIONS - PREPRINTS

- SMILE: Smooth Motion for Improved Long-Horizon VLA Execution** Arxiv, 2026
J Park, E-R Nguyen, K Ranasinghe, C Mata, X Li, M S Ryoo
- LACE: Latent Visual Representation for Cross-Embodiment Learning** Arxiv, 2026
Y S Jang, K Ranasinghe, C Mata, Y Zhang, J Mendez-Mendez, M S Ryoo
- Robotic VLA Benefits from Joint Learning with Motion Image Diffusion** Arxiv, 2025
Y Fang, K Ranasinghe, L Xue, H Zhou, J Tan, R Xu, S Heinecke, C Xiong, S Savarese,
D Szafir, M Ding, M S Ryoo, J C Niebles
- Pixel Motion as Universal Representation for Robot Control** Arxiv, 2025
K Ranasinghe, X Li, C Mata, J Park, M S Ryoo
- Blip-3-Video: You Only Need 32 Tokens to Represent a Video Even in VLMs** Arxiv, 2024
Michael S Ryoo, Honglu Zhou, Shrikant Kendre, Can Qin, Le Xue, Manli Shu, Jongwoo Park,
Kanchana Ranasinghe, Silvio Savarese, Ran Xu, Caiming Xiong, Juan Carlos Niebles
- Test-Time Optimization for Domain Adaptive Open Vocabulary Segmentation** Arxiv, 2025
U Silva, D Samaraweera, S Wanigathunga, K Kariyawasam, K Ranasinghe, M Naseer, R Rodrigo
- LatentCRF: Continuous CRF for Efficient Latent Diffusion** Arxiv, 2024
K Ranasinghe, S Jayasumana, A Veit, A Chakrabarti, D Glasner, M S Ryoo, S Ramalingam, S Kumar

PUBLICATIONS - JOURNAL

- Combined Static & Motion Features for Deep-Networks Based Activity Recognition in Videos**,
S. Ramasinghe, J. Rajasegaran, V. Jayasundara, K Ranasinghe, R. Rodrigo and A. A. Pasqual, IEEE
Transactions on Circuits and Systems for Video Technology, vol. 29, Sept. 2019.

PATENTS

- Kanchana Ranasinghe, Muhammad Muzammal Naseer, Salman Khan, Fahad Khan, **“System and Method for Self-supervised Video Transformer.”** US Patent US20240169692A1 filed on 11/21/2022.

INVITED TALKS

- “Visual Spatial Reasoning with Multimodal LLMs”*, presented (remotely) at Deep Learning Session in Radiology Department, Penn State University, PA (Jan 2025).
- “MVU - an LLM-based framework for solving Long Video Question Answering benchmarks”*, presented (remotely) at Multimodal Weekly organized by Twelve Labs (Aug 2024).
- “Learning to Localize Objects Improves Spatial Reasoning in Visual-LLMs”*, presented at NYC Vision RG, Google Research, NYC (Apr 2024).
- “Perceptual Grouping in Contrastive Vision-Language Models”*, presented at Electronics and Telecommunications Department, University of Moratuwa, Sri Lanka (Jan 2024).
- “Perceptual Grouping in Contrastive Vision-Language Models”*, presented (remotely) at Computer Vision Talks series, ETS Montreal (Oct 2023).

TEACHING

Stony Brook University, NY, USA

- Teaching assistant for CSE 101 course on Introduction to Digital Intelligence
- Teaching assistant for CSE 215 course on Foundations of Computer Science course
- Supporting for graduate level computer vision seminar CSE 656

University of Moratuwa, Sri Lanka

- Supporting for undergraduate level computer vision seminar

SERVICE

Reviewing

- Conference reviewer: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, ICRA 2022 - present
- Journal reviewer: IJCV, TCSVT 2021 - present

Mentoring

- Mentor for undergraduate computer science students at Women in Engineering in Sri Lanka 2023
- Mentor for machine learning competition teams at University of Moratuwa, Sri Lanka 2023
- Mentor undergraduates from University of Moratuwa, Sri Lanka in computer vision research projects and academic writing 2023 - present

OTHER AWARDS

- National Merit Scholarship** - Ranked 13th (2.83 z-score) in Sri Lanka at GCE A/L Examination 2014
- World Rank 296 / National Top 6** - International Mathematical Olympiad (IMO), Colombia 2013